

FINAL PROJECT (CCNA)

Enterprise Network Design & Implementation

Supply Chain Network Documentation

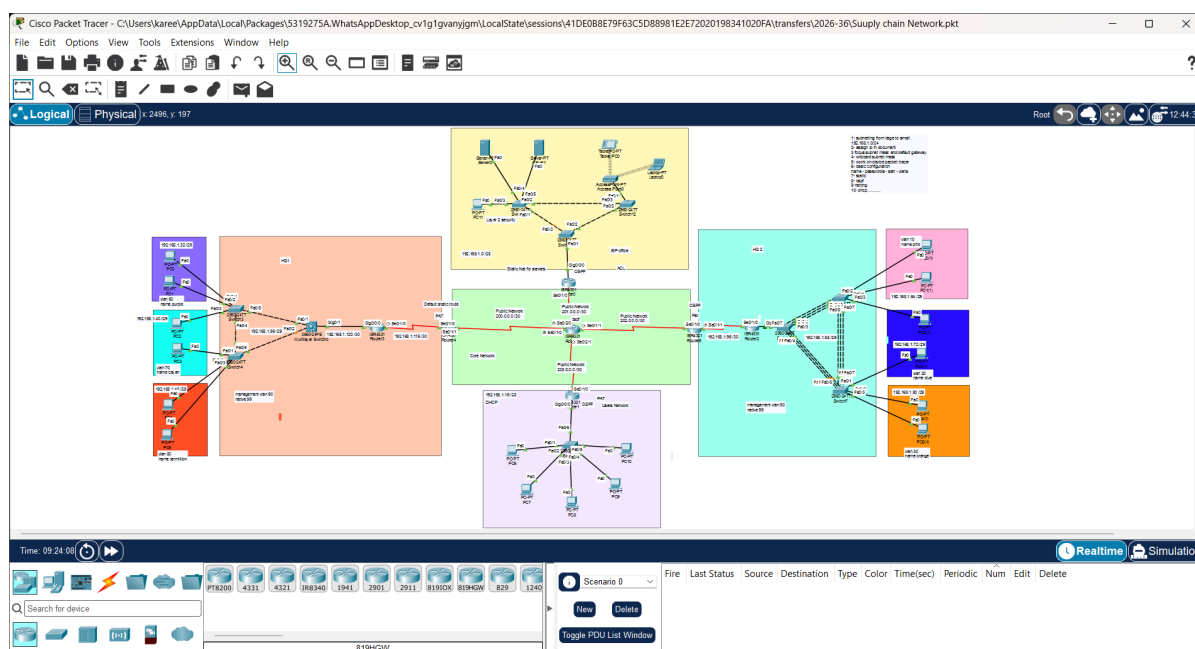


Figure 1 – Complete Logical Topology (HQ1 • HQ2 • Core • ISP Office • Users Network)

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Project	Supply Chain Network – CCNA Final Project
Technologies	VLSM • VLANs • EtherChannel • OSPF • Static Routing • DHCP • PAT / Static NAT • SSH
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1. Project Overview

This document presents the complete design, configuration and verification of a multi-site enterprise network that simulates a Supply Chain environment. The network consists of two Headquarters (HQ1 and HQ2), a Core Network, an ISP Office and a Users Network. All sites are interconnected through serial links and public transit networks.

Design Goals: Hierarchical addressing with VLSM, Layer-2 segmentation using VLANs, link aggregation with EtherChannel (LACP), dynamic routing with OSPF, edge static routing, DHCP for end users, Network Address Translation (PAT + Static NAT for servers), and secure remote management using SSH.

Site	Role	Key Technologies
HQ1	Primary Headquarters	Multilayer Switch, VLANs 60/70/80/90, Static Route, PAT
HQ2	Secondary Headquarters	Router-on-a-stick, VLANs 10/20/30/50, OSPF, PAT, EtherChannel
Core	Central Transit	OSPF Area 0, Public Networks 200/201/202/203
ISP Office	Service Provider Edge	Static NAT (Servers), OSPF, ACL, Layer-2 Security
Users Network	End-User Segment	DHCP, OSPF, PAT

2. Network Topology – Detailed Views

The following figures show the detailed logical topology of each site with interface labels, VLAN assignments and addressing.

2.1 HQ1 Topology

HQ1 is built around a Multilayer Switch (MLS). Four data VLANs (60 purple, 70 cyan, 80 vermillion) and a management VLAN 90 are configured. Native VLAN is 99. The MLS connects to Router3 via a /30 link, which then connects to Router4. Router4 uses a default static route toward the Core and performs PAT.

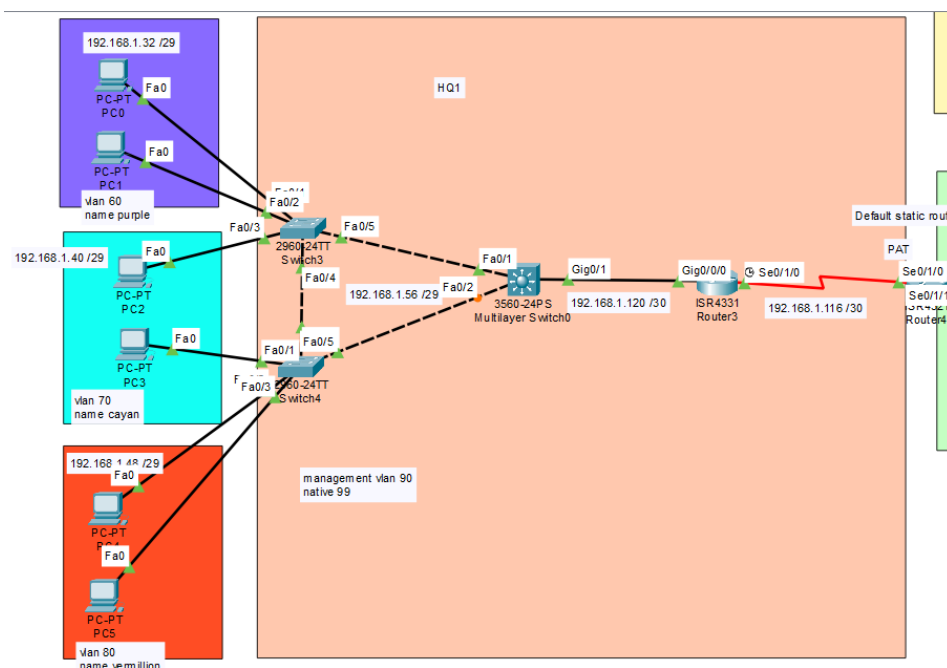


Figure 2 – HQ1 detailed topology (VLANs 60/70/80/90, Multilayer Switch, default static route & PAT on Router4)

2.2 HQ2 Topology

HQ2 uses Router-on-a-stick (Router2) with sub-interfaces for VLANs 10 (pink), 20 (blue), 30 (orange) and management VLAN 50. Native VLAN is 99. EtherChannel (LACP) is implemented between access switches. Router5 connects HQ2 to the Core via a /30 link and participates in OSPF and PAT.

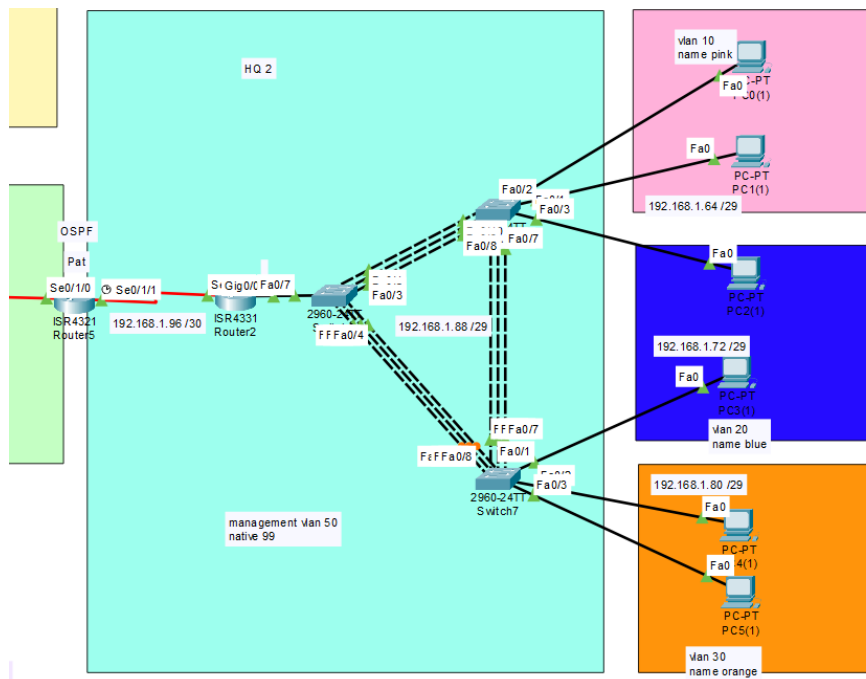


Figure 3 – HQ2 detailed topology (VLANs 10/20/30/50, EtherChannel, OSPF & PAT)

2.3 ISP Office Topology

The ISP Office uses network 192.168.1.0/28. Two servers (Server0 and Server1) are configured with Static NAT for permanent public reachability. Layer-2 security features and an ACL are applied. The site connects to the Core via OSPF.

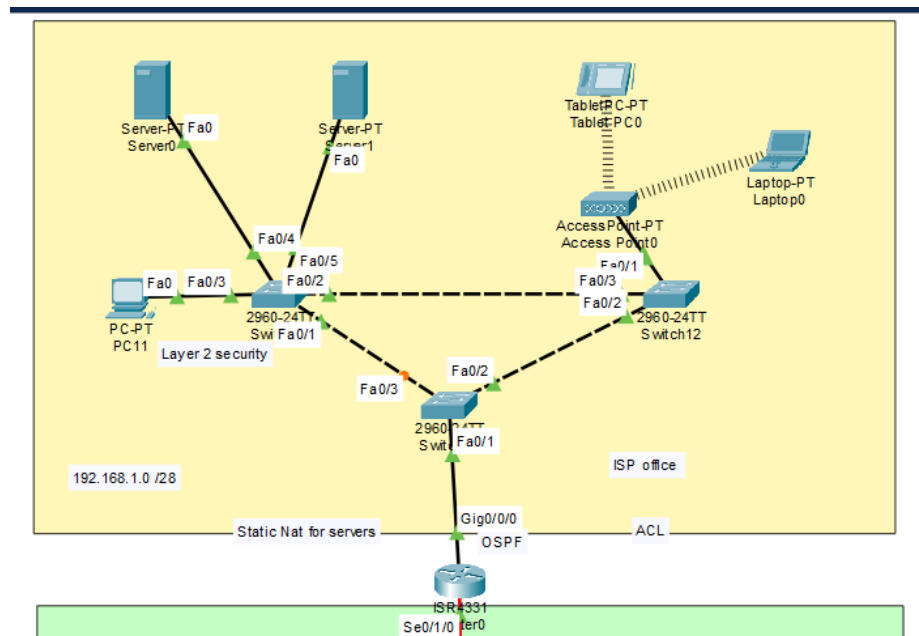


Figure 4 – ISP Office detailed topology (192.168.1.0/28, Static NAT for servers, OSPF, ACL)

2.4 Users Network Topology

The Users Network is a pure access segment (192.168.1.16/28). Router1 acts as the DHCP server and default gateway. Clients PC6–PC10 obtain addresses dynamically. The site participates in OSPF and uses PAT for external access.

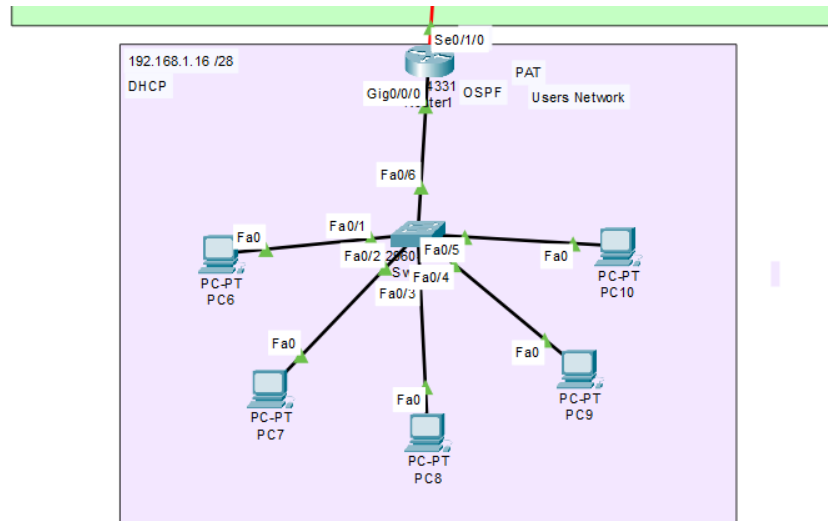


Figure 5 – Users Network detailed topology (DHCP on Router1, OSPF, PAT)

2.5 Core Network Topology

The Core (Router6) is the central transit point. It interconnects all sites using dedicated public networks: 200.0.0.0/30 (HQ1), 201.0.0.0/30 (ISP), 202.0.0.0/30 (HQ2) and 203.0.0.0/30 (Users). OSPF Area 0 runs on all Core interfaces.

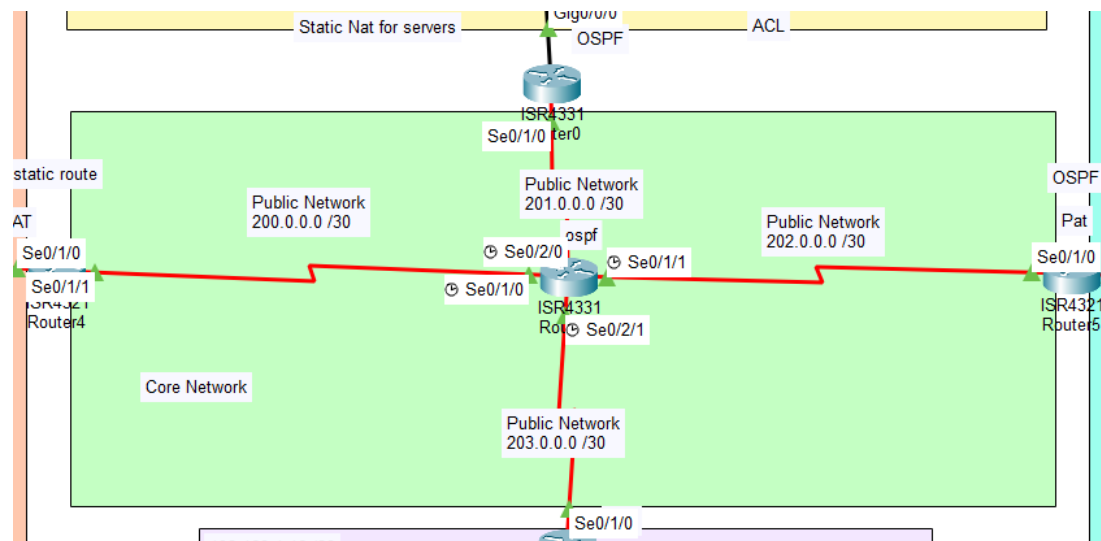


Figure 6 – Core Network detailed topology (Public transit networks & OSPF)

3. Subnetting & IP Addressing Plan

Base Network: 192.168.1.0/24 was subdivided with VLSM to match the exact host requirements of each segment while preserving address space. Serial links received /30 networks. Public transit networks were taken from separate ranges to support NAT and OSPF reachability between sites.

3.1 Subnet Allocation Summary

Purpose	Network	Mask	Hosts	Gateway / SVI
ISP Office	192.168.1.0/28	/28	14	192.168.1.1
Users Network	192.168.1.16/28	/28	14	192.168.1.17
HQ1 VLAN 60 (purple)	192.168.1.32/29	/29	6	192.168.1.33
HQ1 VLAN 70 (cayan)	192.168.1.40/29	/29	6	192.168.1.41
HQ1 VLAN 80 (vermillion)	192.168.1.48/29	/29	6	192.168.1.49
HQ1 VLAN 90 (mgmt)	192.168.1.56/29	/29	6	192.168.1.57
HQ2 VLAN 10 (pink)	192.168.1.64/29	/29	6	192.168.1.65
HQ2 VLAN 20 (blue)	192.168.1.72/29	/29	6	192.168.1.73
HQ2 VLAN 30 (orange)	192.168.1.80/29	/29	6	192.168.1.81
HQ2 VLAN 50 (mgmt)	192.168.1.88/29	/29	6	192.168.1.89
Serial HQ1–R3	192.168.1.120/30	/30	2	—
Serial R3–R4	192.168.1.116/30	/30	2	—
Serial HQ2–R5	192.168.1.96/30	/30	2	—

3.2 Public / Transit Networks

Network	Purpose
200.0.0.0/30	Core ↔ HQ1 (Router4 – Router6)
201.0.0.0/30	Core ↔ ISP Office (Router6 – Router0)
202.0.0.0/30	Core ↔ HQ2 (Router5 – Router6)
203.0.0.0/30	Core ↔ Users Network (Router1 – Router6)

4. Complete Device IP Addressing Tables

Static addressing is used on all devices in HQ1, HQ2 and ISP Office. The Users Network obtains addresses via DHCP from Router1.

4.1 ISP Office

Device / Interface	IP Address	Mask	Gateway
Router0 G0/0/0	192.168.1.1	/28	—
SW10 / SW11 / SW12	192.168.1.2 / .3 / .4	/28	192.168.1.1
PC5	192.168.1.5	/28	192.168.1.1
Server0 / Server1	192.168.1.6 / .7	/28	192.168.1.1
Laptop0 / Tablet0 (via AP)	192.168.1.8 / .9	/28	192.168.1.1
Router0 S0/1/0	201.0.0.4	/30	—

4.2 Users Network (DHCP)

Device / Interface	IP Address	Mask	Gateway
Router1 G0/0/0	192.168.1.17	/28	—
SW9	192.168.1.18	/28	192.168.1.17
PC6 – PC10 (DHCP)	192.168.1.19 – .23	/28	192.168.1.17
Router1 S0/1/0	203.0.0.1	/30	—

4.3 HQ1

Device / Interface	IP Address	Mask	Gateway / Notes
MLS VLAN 60 (purple)	192.168.1.33	/29	SVI
PC0 / PC1 (VLAN 60)	192.168.1.34 / .35	/29	192.168.1.33
MLS VLAN 70 (cayan)	192.168.1.41	/29	SVI
PC2 / PC3 (VLAN 70)	192.168.1.42 / .43	/29	192.168.1.41
MLS VLAN 80 (vermillion)	192.168.1.49	/29	SVI
PC4 / PC5 (VLAN 80)	192.168.1.50 / .51	/29	192.168.1.49
MLS VLAN 90 (mgmt)	192.168.1.57	/29	SVI
SW3 / SW4	192.168.1.59 / .58	/29	192.168.1.57
MLS G0/1	192.168.1.121	/30	Link to R3
Router3 G0/0/0	192.168.1.122	/30	—
Router3 S0/1/0	192.168.1.117	/30	Link to R4
Router4 S0/1/0	192.168.1.118	/30	—
Router4 S0/1/1	200.0.0.1	/30	Link to Core

4.4 HQ2

Device / Interface	IP Address	Mask	Gateway / Notes
Router2 g0/0/0.10 (pink)	192.168.1.65	/29	—
PC0 / PC1 (VLAN 10)	192.168.1.66 / .67	/29	192.168.1.65
Router2 g0/0/0.20 (blue)	192.168.1.73	/29	—
PC2 / PC3 (VLAN 20)	192.168.1.74 / .75	/29	192.168.1.73
Router2 g0/0/0.30 (orange)	192.168.1.81	/29	—
PC4 / PC5 (VLAN 30)	192.168.1.82 / .83	/29	192.168.1.81
Router2 g0/0/0.50 (mgmt)	192.168.1.89	/29	—
SW0 / SW6 / SW7	192.168.1.90 / .91 / .92	/29	192.168.1.89
Router2 S0/1/0	192.168.1.98	/30	Link to R5
Router5 S0/1/1	192.168.1.97	/30	—
Router5 S0/1/0	202.0.0.1	/30	Link to Core

4.5 Core Network (Router6)

Interface	IP Address	Mask	Connected To
S0/1/0	200.0.0.2	/30	HQ1 (Router4)
S0/1/1	202.0.0.2	/30	HQ2 (Router5)
S0/2/0	201.0.0.3	/30	ISP (Router0)
S0/2/1	203.0.0.2	/30	Users (Router1)

5. Implementation Steps

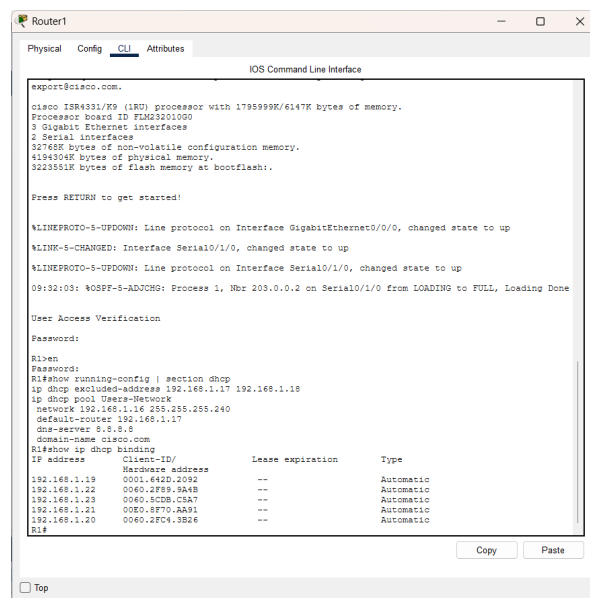
The network was built and configured following the six sequential steps below. Each step includes the technical decision, the configuration approach and the verification evidence.

Step 1 – IP Address Distribution (Static Addressing)

All end devices in HQ1, HQ2 and the ISP Office were assigned static IP addresses according to the VLSM plan. Serial interfaces on every router received the planned /30 addresses. Static addressing guarantees predictable addresses for servers, management interfaces and critical clients, simplifies troubleshooting and removes dependency on DHCP for infrastructure devices.

Step 2 – DHCP Configuration for Users Network

A DHCP pool named **Users-Network** was created on Router1 covering 192.168.1.16/28. The pool excludes the gateway and management addresses and distributes default-router and DNS information. Clients PC6–PC10 automatically obtained addresses.



```

Router1
Physical Config CLI Attributes
IOS Command Line Interface

export@cisco.com:
cisco ISR4331/K9 (18U) processor with 1795999K/6147K bytes of memory.
Processor board ID F1A2301090
3 Gigabit Ethernet interfaces
2 Serial interfaces
32768K bytes of non-volatile configuration memory.
4194504K bytes of physical memory.
3223551K bytes of flash memory at bootflash.

Press RETURN to get started!

%LINEPROTO-5-UPDOWN: Line protocol on Interface GigabitEthernet0/0/0, changed state to up
%LINK-3-CHANGED: Interface Serial0/1/0, changed state to up
%LINEPROTO-5-UPDOWN: Line protocol on Interface Serial0/1/0, changed state to up
09:32:03: %OSPF-5-ADJCHG: Process 1, Nbr 203.0.0.2 on Serial0/1/0 from LOADING to FULL, Loading Done

User Access Verification

Password:
R1>en
Password:
R1#show running-config | section dhcp
ip dhcp excluded-address 192.168.1.17 192.168.1.18
ip dhcp pool Users-Network
network 192.168.1.16 255.255.255.240
default-router 192.168.1.17
dns-server 8.8.8.8
R1#show ip dhcp binding
IP address      Client-ID/
Hardware address Lease expiration Type
192.168.1.19    0001.642D.2092 -- Automatic
192.168.1.22    0060.2F89.8A48 -- Automatic
192.168.1.23    0060.3C3B.C3A7 -- Automatic
192.168.1.21    00E0.8F70.AA91 -- Automatic
192.168.1.20    0060.2FC4.3B26 -- Automatic
R1#
  
```

Figure 7 – DHCP pool configuration and bindings on Router1.

What was done: DHCP pool created; clients received addresses automatically.

Why it was used: DHCP simplifies address management for end-user devices and reduces manual errors.

Step 3 – VLAN & EtherChannel Configuration (HQ1 & HQ2)

HQ1 VLANs: 60 (purple), 70 (cyan), 80 (vermillion), 90 (management), 99 (native).

HQ2 VLANs: 10 (pink), 20 (blue), 30 (orange), 50 (management), 99 (native).

Access ports were assigned to the correct VLANs. Trunks carry all required VLANs with native VLAN 99. EtherChannel (LACP) was configured on Switch0 to increase inter-switch bandwidth and provide link redundancy.

```

Switch0
Physical Config CLI Attributes
IOS Command Line Interface

Switch0#
Switch0#show etherchannel summary
Flags: D - down        P - in port-channel
       I - stand-alone s - suspended
       W - hot-standby (LACP only)
       R - Layer3       S - Layer2
       U - in use       C - failed to allocate aggregator
       u - unsuitable for bundling
       w - waiting to be aggregated
       d - default port

Number of channel-groups in use: 2
Number of aggregators:          2

Group  Port-channel  Protocol    Ports
-----
2      Po2(SU)          LACP       Fa0/4(P) Fa0/5(P) Fa0/6(P)
3      Po3(SU)          LACP       Fa0/1(P) Fa0/2(P) Fa0/3(P)

Switch0#show interfaces trunk
Port      Mode      Encapsulation  Status      Native vlan
Po2       on        802.1q         trunking    99
Po3       on        802.1q         trunking    99
Fa0/7     on        802.1q         trunking    99

Port      Vlans allowed on trunk
Po2       10,20,30,50,60,70,80,99
Po3       10,20,30,50,60,70,80,99
Fa0/7     10,20,30,50,60,70,80,99

Port      Vlans allowed and active in management domain
Po2       10,20,30,50,60,70,80,99
Po3       10,20,30,50,60,70,80,99
Fa0/7     10,20,30,50,60,70,80,99

Port      Vlans in spanning tree forwarding state and not pruned
Po2       10,20,30,50,60,70,80,99
Po3       10,20,30,50,60,70,80,99
Fa0/7     10,20,30,50,60,70,80,99

Switch0#
  
```

Figure 8 – show etherchannel summary & show interfaces trunk on SW0.

What was done: LACP EtherChannel formed (Po2 & Po3); required VLANs allowed on trunks.

Why it was used: EtherChannel increases bandwidth and provides redundancy between access and distribution layers.

Step 4 – Routing Configuration (OSPF + Static)

OSPF Area 0 was enabled on all routers **except HQ1**. On Router4 (HQ1 edge) a default static route toward the Core was configured instead. This keeps the HQ1 edge simple and stable while still providing full reachability through the Core. OSPF neighbors reached the FULL state and all remote subnets appeared correctly in the routing tables.

```

Router6
Physical Config CLI Attributes
IOS Command Line Interface

%LINEPROTO-5-UPDOWN: Line protocol on Interface Serial0/1/1, changed state to down
09:05:04: %OSPF-5-ADJCHG: Process 1, Nbr 202.0.0.1 on Serial0/1/1 from FULL to DOWN, Neighbor Down:
Interface down or detached
%LINEPROTO-5-UPDOWN: Line protocol on Interface Serial0/2/1, changed state to down
09:05:04: %OSPF-5-ADJCHG: Process 1, Nbr 203.0.0.1 on Serial0/2/1 from FULL to DOWN, Neighbor Down:
Interface down or detached
%LINEPROTO-5-UPDOWN: Line protocol on Interface Serial0/1/1, changed state to up
%LINEPROTO-5-UPDOWN: Line protocol on Interface Serial0/2/0, changed state to up
%LINEPROTO-5-UPDOWN: Line protocol on Interface Serial0/2/1, changed state to up
09:05:14: %OSPF-5-ADJCHG: Process 1, Nbr 203.0.0.1 on Serial0/2/1 from LOADING to FULL, Loading Done
09:05:15: %OSPF-5-ADJCHG: Process 1, Nbr 202.0.0.1 on Serial0/1/1 from LOADING to FULL, Loading Done

User Access Verification
Password:
R6>en
Password:
R6#show ip ospf neighbor

Neighbor ID  Pri  State       Dead Time   Address        Interface
202.0.0.1    0    FULL/  -    00:00:35    202.0.0.1      Serial0/1/1
203.0.0.1    0    FULL/  -    00:00:39    203.0.0.1      Serial0/2/1

R6#show ip route ospf

O 192.168.1.0/24 is variably subnetted, 6 subnets, 3 masks
O   192.168.1.6 [110/65] via 203.0.0.1, 00:23:41, Serial0/2/1
O   192.168.1.64 [110/129] via 202.0.0.1, 00:23:41, Serial0/1/1
O   192.168.1.72 [110/129] via 202.0.0.1, 00:23:41, Serial0/1/1
O   192.168.1.80 [110/129] via 202.0.0.1, 00:23:41, Serial0/1/1
O   192.168.1.88 [110/129] via 202.0.0.1, 00:23:41, Serial0/1/1
O   192.168.1.96 [110/129] via 202.0.0.1, 00:23:41, Serial0/1/1
O*E2 0.0.0.0/0 [110/1] via 202.0.0.1, 00:23:41, Serial0/1/1

R6#
  
```

Figure 9 – show ip ospf neighbor & show ip route ospf on Router6.

What was done: OSPF Area 0 enabled; full adjacency established with 202.0.0.1 and 203.0.0.1.
Why it was used: OSPF provides dynamic routing that automatically adapts to topology changes.

```

Router4
Physical Config CLI Attributes
IOS Command Line Interface

A summary of U.S. laws governing Cisco cryptographic products may be found at:
http://www.cisco.com/wml/export/cryptoc/tool/stqrg.html

If you require further assistance please contact us by sending email to
export@cisco.com.

Cisco ISR4321/K9 (1RU) processor with 1687137K/6147K bytes of memory.
Processor board ID FLM2041W2BD
2 Gigabit Ethernet interfaces
2 Serial interfaces
32768K bytes of non-volatile configuration memory.
4194304K bytes of physical memory.
3223551K bytes of flash memory at bootflash.

Press RETURN to get started!

%LINK-5-CHANGED: Interface Serial0/1/0, changed state to up
%LINK-5-CHANGED: Interface Serial0/1/1, changed state to up
%LINEPROTO-5-UPDOWN: Line protocol on Interface Serial0/1/1, changed state to up
%LINEPROTO-5-UPDOWN: Line protocol on Interface Serial0/1/0, changed state to up

User Access Verification
Password:
R4>en
Password:
R4#show ip route static

S 192.168.1.0/24 is variably subnetted, 8 subnets, 4 masks
S   192.168.1.0/25 is directly connected, Serial0/1/0
S   192.168.1.32/29 is directly connected, Serial0/1/0
S   192.168.1.40/29 [1/0] via 192.168.1.117
S   192.168.1.48/29 [1/0] via 192.168.1.117
S   192.168.1.56/29 [1/0] via 192.168.1.117
S   192.168.1.64/30 [1/0] via 192.168.1.117
S*  0.0.0.0/0 is directly connected, Serial0/1/1

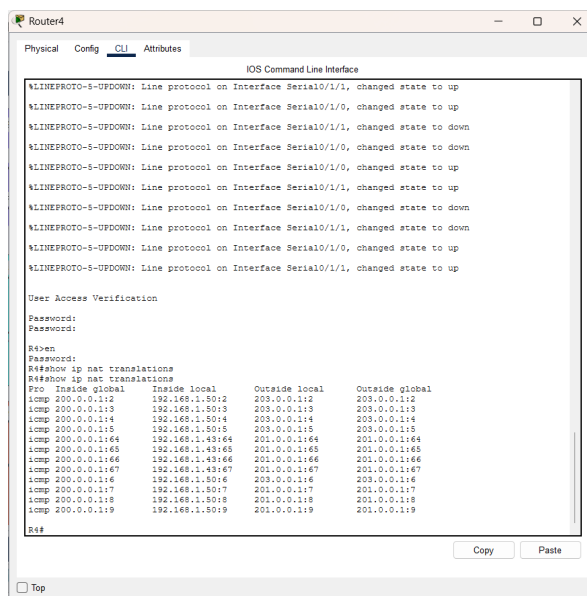
R4#
  
```

Figure 10 – show ip route static on Router4 (HQ1 edge).

What was done: Static routes for HQ1 internal networks plus a default route toward the Core.
Why it was used: Static routing on the HQ1 edge keeps the configuration simple where dynamic updates are not required.

Step 5 – NAT Configuration (PAT + Static NAT)

PAT (NAT Overload) was configured on the edge routers so that all private hosts can reach the public/ISP side. The two servers in the ISP Office (Server0 and Server1) received **Static NAT** entries to guarantee permanent public reachability. All other internal networks use dynamic PAT.



The screenshot shows the CLI of Router4 with the following output:

```

IOS Command Line Interface
%LINEPROTO-5-UPDOWN: Line protocol on Interface Serial0/1/1, changed state to up
%LINEPROTO-5-UPDOWN: Line protocol on Interface Serial0/1/0, changed state to up
%LINEPROTO-5-UPDOWN: Line protocol on Interface Serial0/1/1, changed state to down
%LINEPROTO-5-UPDOWN: Line protocol on Interface Serial0/1/0, changed state to down
%LINEPROTO-5-UPDOWN: Line protocol on Interface Serial0/1/0, changed state to up
%LINEPROTO-5-UPDOWN: Line protocol on Interface Serial0/1/1, changed state to up
%LINEPROTO-5-UPDOWN: Line protocol on Interface Serial0/1/0, changed state to down
%LINEPROTO-5-UPDOWN: Line protocol on Interface Serial0/1/1, changed state to down
%LINEPROTO-5-UPDOWN: Line protocol on Interface Serial0/1/0, changed state to up
%LINEPROTO-5-UPDOWN: Line protocol on Interface Serial0/1/1, changed state to up

User Access Verification
Password:
Password:

R#en
Password:
R#show ip nat translations
R#show ip nat translations

```

Pro	Inside global	Inside local	Outside local	Outside global
icmp	200.0.0.112	192.168.1.5012	203.0.0.112	203.0.0.112
icmp	200.0.0.113	192.168.1.5013	203.0.0.113	203.0.0.113
icmp	200.0.0.114	192.168.1.5014	203.0.0.114	203.0.0.114
icmp	200.0.0.115	192.168.1.5015	203.0.0.115	203.0.0.115
icmp	200.0.0.116	192.168.1.43164	201.0.0.116	201.0.0.116
icmp	200.0.0.1165	192.168.1.43165	201.0.0.1165	201.0.0.1165
icmp	200.0.0.1166	192.168.1.43166	201.0.0.1166	201.0.0.1166
icmp	200.0.0.1167	192.168.1.43167	201.0.0.1167	201.0.0.1167
icmp	200.0.0.116	192.168.1.5016	203.0.0.116	203.0.0.116
icmp	200.0.0.117	192.168.1.5017	201.0.0.117	201.0.0.117
icmp	200.0.0.118	192.168.1.5018	201.0.0.118	201.0.0.118
icmp	200.0.0.119	192.168.1.5019	201.0.0.119	201.0.0.119

R#

Figure 11 – show ip nat translations on Router4.

What was done: PAT applied so private HQ1 hosts can reach external networks.

Why it was used: PAT conserves public IP addresses by allowing many internal hosts to share one public address.

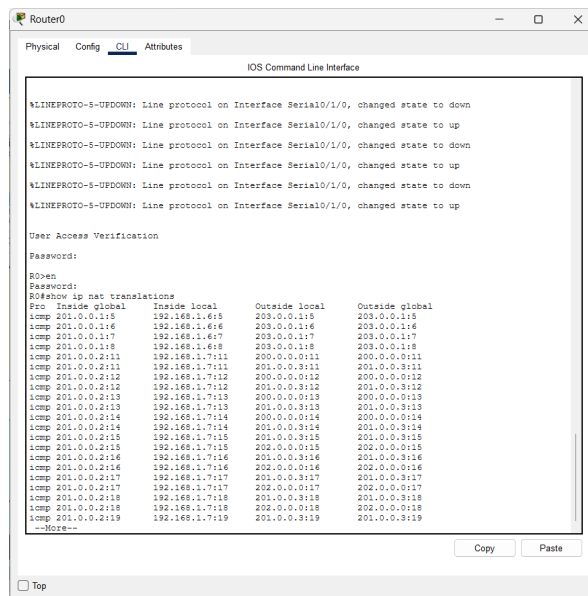


Figure 12 – show ip nat translations on Router0 (ISP side).

What was done: PAT verified on the ISP edge; Static NAT configured for the two servers.

Why it was used: Static NAT gives servers permanent public reachability while PAT handles the rest of the hosts.

Step 6 – Device Security (Passwords, SSH & Telnet)

Console, VTY and enable-secret passwords were configured on every router and switch. SSH was preferred over Telnet for encrypted management access. A domain name was set, RSA keys were generated, and VTY lines were restricted to SSH (or both SSH and Telnet where required). Successful SSH login was verified from an end device to the Multilayer Switch.

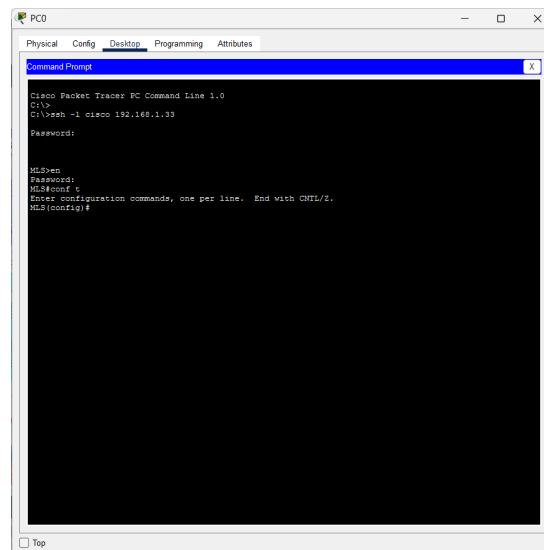


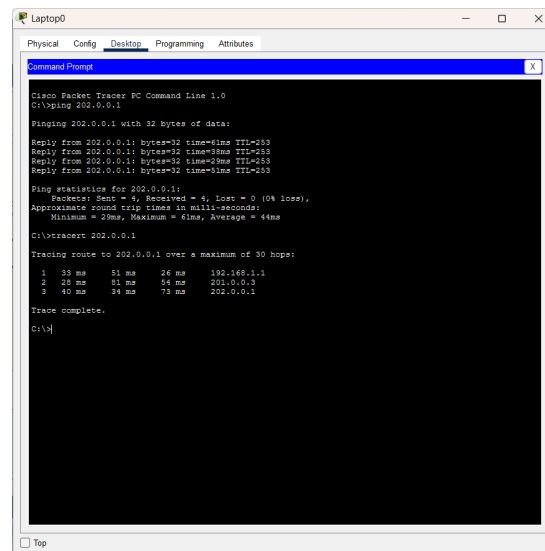
Figure 13 – Successful SSH login from PC0 to the Multilayer Switch (192.168.1.33).

What was done: SSH configured and remote login successfully performed.

Why it was used: SSH encrypts management traffic and is preferred over Telnet for secure administration.

6. End-to-End Connectivity Verification

From Laptop0 in the ISP Office a ping and traceroute were executed toward the Core side of the HQ2 link. Zero packet loss and a correct multi-hop path confirm full reachability and proper routing, NAT and addressing across the entire network.



```
Laptop0
Physical Config Desktop Programming Attributes
Command Prompt
Cisco Packet Tracer PC Command Line 1.0
C:\>ping 202.0.0.1

Pinging 202.0.0.1 with 32 bytes of data:
Reply from 202.0.0.1: bytes=32 time=4ms TTL=255
Reply from 202.0.0.1: bytes=32 time=3ms TTL=255
Reply from 202.0.0.1: bytes=32 time=2ms TTL=255
Reply from 202.0.0.1: bytes=32 time=3ms TTL=255

Ping statistics for 202.0.0.1:
    Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
    Approximate round trip times in milli-seconds:
        Minimum = 2ms, Maximum = 6ms, Average = 4ms

C:\>tracert 202.0.0.1

Tracing route to 202.0.0.1 over a maximum of 30 hops:
  0  33 ms  51 ms  26 ms  192.168.1.1
  1  28 ms  51 ms  56 ms  202.0.0.0
  2  40 ms  34 ms  73 ms  202.0.0.1

Trace complete.

C:\>
```

Figure 14 – Ping to 202.0.0.1 succeeds (0 % loss). Traceroute shows the correct path.

What was done: End-to-end connectivity tested with ping and traceroute from the ISP Office.

Why it was used: These tests confirm that routing, NAT and addressing work correctly across the whole network.

7. Summary of Work Completed

The following tasks were successfully completed and verified:

1. **IP Address Distribution** – Static addressing applied to all devices in HQ1, HQ2 and ISP Office according to the VLSM plan.
2. **DHCP** – DHCP pool configured on Router1 for the Users Network; clients automatically received IP addresses, gateway and DNS.
3. **VLANs & EtherChannel** – VLANs created for HQ1 (60, 70, 80, 90) and HQ2 (10, 20, 30, 50) with proper trunking and LACP EtherChannel.
4. **Routing** – OSPF Area 0 configured on all routers except HQ1, which uses a default static route toward the Core.
5. **NAT** – PAT (NAT Overload) applied for all internal networks; Static NAT configured specifically for the two servers in the ISP Office.
6. **Security** – Passwords, SSH and Telnet configured on all routers and switches; remote SSH access verified.

All verification commands (show commands, ping and traceroute) confirm that the network is fully operational and ready for presentation or publication.

8. Conclusion

The Final_Project (CCNA) successfully implements a multi-site enterprise network with hierarchical design, VLSM subnetting, Layer-2 redundancy (EtherChannel), secure device management (SSH), dynamic routing (OSPF), edge static routing, and Network Address Translation (PAT + Static NAT). All required verification steps confirm that the network is complete, stable and fully functional.

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